EDA Report

Exploratory Data Analysis Report: A2_data_release.csv

> Target column: `Appliances` (appliance energy consumption, in Wh). This report only performs analysis and visualization, with no modeling.

1. Data overview

Item	Result
Sampling interval	**10 minutes** (all 999 adjacent time differences are `0 days 00:10:00`, strictly evenly spaced)
Start time	**2016-01-11 17:00:00**
End time	**2016-01-18 15:30:00**
Time span	6 days 22 hours 30 minutes (≈ 6.94 days)
Rows × columns	**1000 rows × 29 columns**
Missing values	**None** (all 28 numeric columns have 0 missing values)

Column composition: 1 timestamp column `date` + 28 numeric features

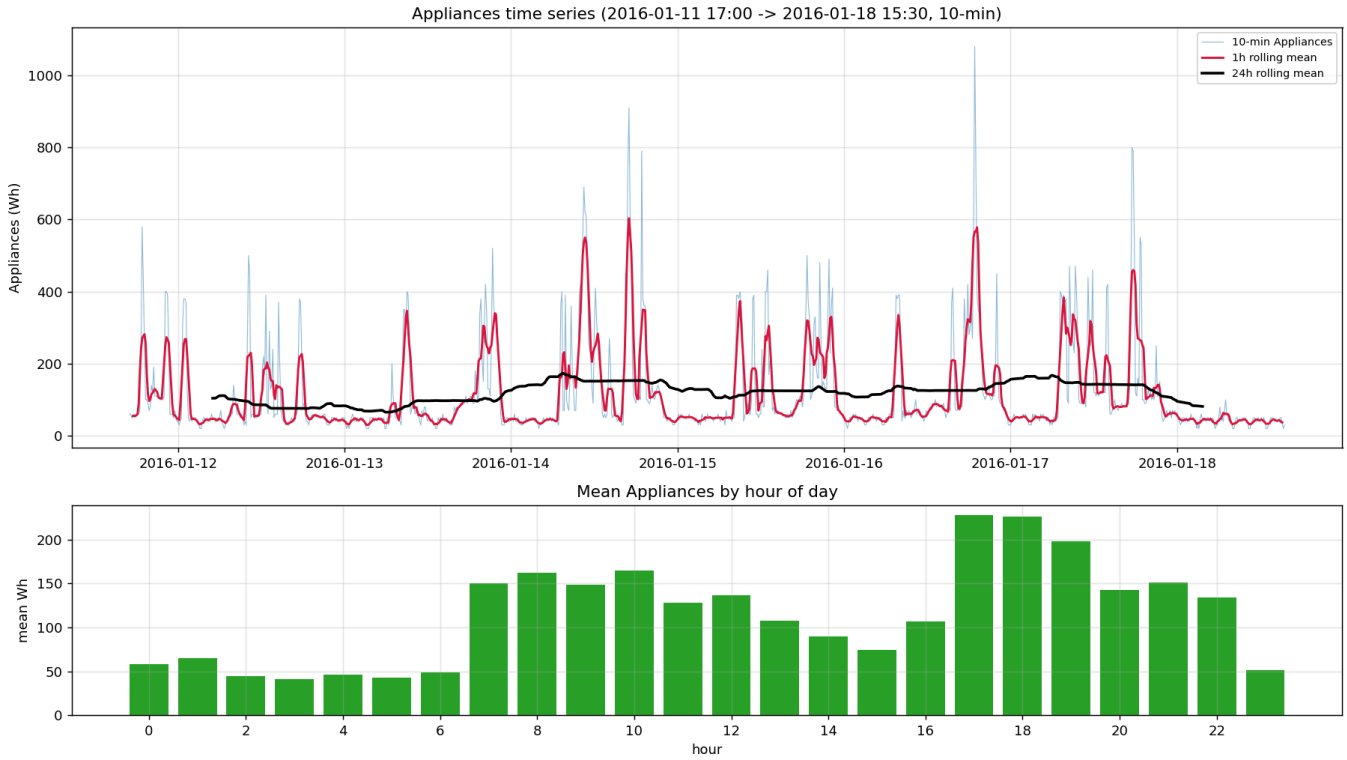
- Target: `Appliances`
- Others: `lights`; 9 groups of indoor temperature/humidity `T1–T9` / `RH_1–RH_9`; outdoor `T_out`, `RH_out`, `Press_mm_hg`, `Windspeed`, `Visibility`, `Tdewpoint`; random variables `rv1`, `rv2`.

2. Statistics of each column

Column	min	mean	max	std
Appliances	20	114.46	1080	131.388
lights	0	6.19	70	10.1137
T1	18.39	20.8896	23.79	1.4007
RH_1	32.0233	41.6689	55.8933	2.9115
T2	17.7	20.0262	22.79	1.3598
RH_2	30.16	40.5855	47.2233	2.5577
T3	19.1333	20.5396	22.79	0.8173
RH_3	35.03	42.6305	48.53	2.1667
T4	18	19.7468	22.39	1.0037
RH_4	35.5	42.0362	49.9475	2.6643
T5	17	18.3768	21.9967	0.9909
RH_5	41.5	55.2272	91.2933	8.5669
T6	-5.535	2.4559	7.9	3.1579
RH_6	51.8233	88.4333	97.3	9.5441
T7	17.1	18.6445	21.79	1.0029
RH_7	30.736	39.6495	46.2633	3.2744
T8	17.79	19.5871	22.76	1.149
RH_8	37.2225	47.0823	55.09	3.5544
T9	16.6	17.3295	18.5	0.4351
RH_9	36.43	43.9324	50.4	3.2848
T_out	-4.8	2.7117	7.5	2.8078
Press_mm_hg	733.5	754.631	765.6	8.6105
RH_out	59	86.2365	100	8.5934
Windspeed	1	4.969	11	2.3244
Visibility	14	35.148	65	9.2176
Tdewpoint	-6.5	0.5323	5.3	2.9236
rv1	0.0657	24.9846	49.9965	14.4194
rv2	0.0657	24.9846	49.9965	14.4194

Key points: `Appliances` is highly volatile (std≈131, about 1.15× its mean; max 1080 far above the mean), clearly right-skewed/peaked; `rv1` and `rv2` have exactly identical values and are uninformative random/redundant columns.

3. Target series plot



The figure shows the 10-minute raw series (light blue) overlaid with a 1-hour rolling mean (red) and a 24-hour rolling mean (black), with a bar chart of the mean aggregated by hour of day below.

4. Intraday periodicity assessment

Conclusion: there is a clear intraday cycle, dominated by two components of about 12 hours (semi-daily) and about 24 hours (daily).

Basis:

1. **The autocorrelation (ACF) shows local peaks around about 1 day and its multiples**: lag=144 (24h)≈**0.128**, lag=288 (48h)≈**0.169**. Apart from the very-short-lag correlation (lag=1≈0.77), these daily-recurrence peaks are the most structured periodic signals in the series.

   > Note: measured ACF at lag=432 (72h) is only ≈0.089 and does not form a peak, so the claim that "peaks fall strictly at integer multiples of 24h" does not hold; limited by only about 7 days of data, higher-order autocorrelation is itself unstable. The only truly stable peaks are at 24h and 48h.

2. **The spectral (periodogram) main peak corresponds to about 12 hours and the secondary peak to about 24 hours**: the maximum power peak period is ≈ **11.9 h** (power ≈ 7.65×10^{8}), followed by ≈ **23.8 h** (power ≈ 5.77×10^{8}), exhibiting a typical day-night structure dominated by the semi-daily harmonic with the daily period as a secondary component.

3. **The mean aggregated by hour shows a clear day-night pattern**: lowest in the early hours 0–6 and at 23 (mean ≈ 41–65 Wh), with peaks in the morning and evening — 7–10 a.m. ≈ 149–165 Wh, and a peak in the evening 17–19 ≈ 198–228 Wh. The standard deviation of the hourly means is about 0.45 of the overall standard deviation, indicating stable systematic intraday variation.

Combining the three lines of evidence, `Appliances` has a significant intraday cycle: in the spectrum the semi-daily (≈11.9h) component is strongest and the daily (≈23.8h) component second; the autocorrelation shows local peaks at 24h/48h; and the hourly-aggregated means show a stable daily day-night pattern (peak in the evening 17–18, trough in the early hours 3–4).